413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 9

Prof. Suryanarayana Maddu
Department of Applied Mathematics and Statistics
Data Science and AI Institute
Johns Hopkins University

The **method of moments** uses the moments of the distribution,

$$\mu_k = \mathbb{E}[X^k],$$

to estimate the parameters θ of a probability distribution function. If the parameter can be written in terms of the population moments,

$$\theta = g(\mu_1, \mu_2, \ldots),$$

we estimate it by replacing the population moments with their sample counterparts:

$$\boxed{\hat{\theta}_{\text{MM}} = g(\hat{\mu}_1, \hat{\mu}_2, \ldots).}$$

where,

$$\hat{\mu}_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k.$$

1 Poisson Distribution

Suppose

$$X_i \sim \text{Poisson}(\lambda),$$

where λ is an unknown parameter, but we know the structure of the distribution (Poisson). For a Poisson random variable,

$$\mathbb{E}[X] = \lambda.$$

Therefore,

$$\lambda = \mu_1.$$

Replacing the population moment by the sample moment gives

$$\boxed{\hat{\lambda}_{\text{MM}} = \hat{\mu}_1 = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}_n.}$$

Thus, the method-of-moments estimate of the Poisson rate is simply the sample mean.

1.1 Sampling variability of $\widehat{\lambda}_{\text{MM}}$

The estimator $\widehat{\lambda}_{\text{MM}} = \overline{X}_n$ is itself a random variable because it depends on the random sample $X_1, \ldots, X_n$. Its expectation is

$$E[\widehat{\lambda}_{\text{MM}}] = E[\overline{X}_n] = \lambda,$$

so $\overline{X}_n$ is unbiased for λ. Also,

$$\text{Var}(\widehat{\lambda}_{\text{MM}}) = \text{Var}(\overline{X}_n) = \text{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\,\text{Var}\left(\sum_{i=1}^{n} X_i\right).$$

For a Poisson random variable $\text{Var}(X) = \lambda$, so

$$\text{Var}(\widehat{\lambda}_{\text{MM}}) = \text{Var}(\overline{X}_n) = \frac{\lambda}{n}.$$

Therefore,

$$\boxed{\mathrm{SE}(\widehat{\lambda}_{\mathrm{MM}}) = \sqrt{\frac{\lambda}{n}}.}$$

The standard error depends on the unknown λ. We therefore use a plug-in estimate:

$$\boxed{\widehat{\mathrm{SE}}(\widehat{\lambda}_{\mathrm{MM}}) = \sqrt{\frac{\widehat{\lambda}_{\mathrm{MM}}}{n}} = \sqrt{\frac{\overline{X}_n}{n}}.}$$

2 Gaussian mean and variance

Suppose

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2).$$

There are two unknown parameters:

$$\theta = (\mu, \sigma^2).$$

We therefore use two moment equations. The first and second population moments are

$$\mu_1 = E[X] = \mu, \quad \mu_2 = E[X^2].$$

The corresponding sample moments are

$$\widehat{\mu}_1 = \overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \quad \widehat{\mu}_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2.$$

Using the identity

$$\mathrm{Var}(X) = E[X^2] - \{E[X]\}^2$$

we can write variance σ^2 as a function of the first and second moment,

$$\sigma^2 = \mu_2 - \mu_1^2.$$

Substituting the population moments with sample moments, we can arrive at the estimates of the parameters.

$$\widehat{\mu}_{\mathrm{MM}} = \overline{X}_n, \quad \widehat{\sigma}^2_{\mathrm{MM}} = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \overline{X}_n^2.$$

Using the identity

$$\sum_{i=1}^{n}(X_i - \overline{X})^2 = \sum_{i=1}^{n} X_i^2 - n\overline{X}^2,$$

we can write the variance estimator as

$$\boxed{\widehat{\sigma}^2_{\mathrm{MM}} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2.}$$

This is the method-of-moments estimator of σ^2. It is not the usual *unbiased* sample variance, which is

$$S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

The two estimators are related by

$$\widehat{\sigma}^2_{\mathrm{MM}} = \frac{n-1}{n} S_n^2.$$

2.1 Sampling variability of the Gaussian mean estimator $\widehat{\mu}_{\mathrm{MM}}$

Since

$$X_i \sim \mathcal{N}(\mu, \sigma^2),$$

the sample mean has an exact Gaussian sampling distribution:

$$\overline{X}_n \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

Therefore,

$$\mathrm{Var}(\widehat{\mu}_{\mathrm{MM}}) = \frac{\sigma^2}{n}, \quad \mathrm{SE}(\widehat{\mu}_{\mathrm{MM}}) = \frac{\sigma}{\sqrt{n}}.$$

Since σ is usually unknown, a standard plug-in estimate is

$$\widehat{\mathrm{SE}}(\widehat{\mu}_{\mathrm{MM}}) = \frac{S_n}{\sqrt{n}}.$$

2.2 Sampling variability of the Gaussian variance estimator $\widehat{\sigma}^2_{\mathrm{MM}}$

For Gaussian data,

$$\frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2_{n-1} \qquad \text{(proof not shown)}.$$

Hence,

$$\mathrm{Var}(S_n^2) = \frac{2\sigma^4}{n-1}, \qquad \mathrm{SE}(S_n^2) = \sigma^2\sqrt{\frac{2}{n-1}}.$$

For the method-of-moments estimator

$$\widehat{\sigma}^2_{\mathrm{MM}} = \frac{n-1}{n}S_n^2,$$

we obtain

$$\mathrm{Var}\big(\widehat{\sigma}^2_{\mathrm{MM}}\big) = \frac{2\sigma^4(n-1)}{n^2}, \qquad \mathrm{SE}\big(\widehat{\sigma}^2_{\mathrm{MM}}\big) = \sigma^2\frac{\sqrt{2(n-1)}}{n}.$$

Using $\widehat{\sigma}^2_{\mathrm{MM}}$ as a plug-in estimate of σ^2,

$$\boxed{\widehat{\mathrm{SE}}\big(\widehat{\sigma}^2_{\mathrm{MM}}\big) = \widehat{\sigma}^2_{\mathrm{MM}}\frac{\sqrt{2(n-1)}}{n}.}$$

For Gaussian data, these quantities can be derived analytically because the sampling distribution of the variance estimator is known exactly. In more complicated models, however, the sampling distribution, and therefore the analytical variance or standard error of an estimator is often difficult or impossible to derive in closed form.

2.3 Why standard errors become difficult

For the Poisson mean and Gaussian mean, the estimator is a sample average:

$$\widehat{\theta} = \overline{X}_n.$$

Its variability follows directly from

$$\mathrm{Var}(\overline{X}_n) = \frac{\mathrm{Var}(X)}{n}.$$

For the Gaussian variance estimator, an exact answer is available (see previous section) because of a special property of Gaussian samples:

$$\frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2_{n-1}.$$

However, an estimator may be much more complicated. For example, it may be:

- a sample quantile, such as the median;
- a ratio of two estimated quantities;

- the solution to a nonlinear system of equations;
- a regression coefficient in a complex model;
- an estimator produced by an iterative optimization algorithm;
- a statistic after model selection or data preprocessing.

In such cases, we still want to answer the same question:

If we repeated the experiment many times, how much would the estimator vary?

More specifically, how can we approximate the variance or the shape of the distribution of

$$\widehat{\theta} = \widehat{\theta}(X_1, \ldots, X_n).$$

in such cases. Bootstrap methods help us approximate the distribution of the estimator numerically through simulations.

2.4 Parametric bootstrap

Suppose we specify a parametric model

$$p_\theta(x)$$

and estimate its parameter from the observed data (using method of moments or any method):

$$\widehat{\theta} = \widehat{\theta}(x_1, \ldots, x_n).$$

The parametric bootstrap uses the fitted model

$$p_{\widehat{\theta}}(x)$$

as an approximation to the unknown data-generating distribution.

The central idea is:

Instead of repeating the real experiment many times, simulate repeated experiments from the fitted model.

For bootstrap replicate $b = 1, \ldots, B$:

1. Sample a new dataset of size n from $p_{\widehat{\theta}}(x)$:

$$X_1^{*(b)}, \ldots, X_n^{*(b)} \overset{\text{iid}}{\sim} p_{\widehat{\theta}}.$$

2. Apply exactly the same estimation procedure to the bootstrap sample:

$$\widehat{\theta}^{*(b)} = \widehat{\theta}\left(X_1^{*(b)}, \ldots, X_n^{*(b)}\right).$$

The resulting estimators

$$\widehat{\theta}^{*(1)}, \ldots, \widehat{\theta}^{*(B)}$$

approximate the sampling distribution of $\widehat{\theta}$. The parametric bootstrap standard error is the sample standard deviation of the bootstrap estimates:

$$\boxed{\widehat{\mathrm{SE}}_{\mathrm{boot}}(\widehat{\theta}) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\widehat{\theta}^{*(b)} - \overline{\widehat{\theta}^{*}}\right)^2},}$$

where

$$\overline{\widehat{\theta}^{*}} = \frac{1}{B}\sum_{b=1}^{B}\widehat{\theta}^{*(b)}.$$

2.5 Example: parametric bootstrap for Gaussian variance

Suppose the observed data are modeled as

$$X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2).$$

From the observed data $X_1, \ldots, X_n$, compute

$$\widehat{\mu} = \overline{X}_n, \quad S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2.$$

For each bootstrap replicate b, simulate

$$X_1^*, \ldots, X_n^* \overset{\text{iid}}{\sim} \mathcal{N}(\widehat{\mu}_{\text{MM}}, \widehat{\sigma}^2_{\text{MM}}),$$

then estimate

$$\widehat{\mu}^* = \overline{X}_n^*, \quad S_n^{2*} = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i^* - \overline{X}_n^*\right)^2.$$

The standard deviation of

$$S^{2*(1)}, \ldots, S^{2*(B)}$$

estimates the repeated-sampling standard error of S_n^2. For Gaussian data, we also have the analytic result

$$\text{SE}(S_n^2) = \sigma^2\sqrt{\frac{2}{n-1}}.$$

The plug-in analytic estimate is

$$S_n^2\sqrt{\frac{2}{n-1}}.$$

Thus the Gaussian variance example is useful because it lets us compare the analytic standard error with the parametric bootstrap standard error. In more complex settings, the bootstrap may be available even when an exact analytic derivation is not.

The method of moments estimates unknown parameters by matching model-implied moments to sample moments:

$$E_\theta[X^k] \quad \longleftrightarrow \quad \frac{1}{n}\sum_{i=1}^{n} X_i^k.$$

For simple models, we can derive the sampling variability of estimators analytically. For example,

$$\text{SE}(\overline{X}) = \sqrt{\frac{\lambda}{n}} \qquad \text{for } X_i \sim \text{Poisson}(\lambda),$$

and

$$\text{SE}(S_n^2) = \sigma^2\sqrt{\frac{2}{n-1}} \qquad \text{for Gaussian data.}$$

When the sampling distribution of an estimator is difficult to derive, parametric bootstrap methods provide a computational approximation:

2.6 Nonparametric bootstrap

The parametric bootstrap assumes that the fitted parametric model is an adequate approximation to the data-generating distribution. For example, a Gaussian parametric bootstrap assumes that new datasets should be simulated from

$$\mathcal{N}(\widehat{\mu}_{\text{MM}}, \widehat{\sigma}^2_{\text{MM}}).$$

The nonparametric bootstrap does not specify a parametric family such as Gaussian, Poisson, or Bernoulli. Instead, it uses the empirical distribution of the observed data.

Given observed values

$$X_1, \ldots, X_n,$$

the empirical distribution assigns probability $\frac{1}{n}$ to each observed value. For bootstrap replicate b, draw

$$X_1^{*(b)}, \ldots, X_n^{*(b)}$$

by sampling from

$$\{X_1, \ldots, X_n\}$$

with replacement. Then recompute the estimator:

$$\widehat{\theta}^{*(b)} = \widehat{\theta}\left(X_1^{*(b)}, \ldots, X_n^{*(b)}\right).$$

The nonparametric bootstrap standard error is again

$$\boxed{\widehat{\mathrm{SE}}_{\mathrm{boot}}(\widehat{\theta}) = \mathrm{SD}\left(\widehat{\theta}^{*(1)}, \ldots, \widehat{\theta}^{*(B)}\right).}$$

Parametric versus nonparametric bootstrap

	Parametric bootstrap	**Nonparametric bootstrap**
Resampling mechanism	Simulate from $p_{\widehat{\theta}}$	Resample observed values
Model assumption	Assumes a parametric model	Uses the empirical distribution
Gaussian example	$X_i^* \sim \mathcal{N}(\widehat{\mu}_{\mathrm{MM}}, \widehat{\sigma}^2_{\mathrm{MM}})$	X_i^* sampled from $\{X_1, \ldots, X_n\}$
Poisson example	$X_i^* \sim \mathrm{Poisson}(\widehat{\lambda}_{\mathrm{MM}})$	X_i^* sampled from observed counts

2.7 Maximum Likelihood Estimation

The method of moments estimates parameters by matching a few model moments to their empirical counterparts. For example,

$$\mathbb{E}_\theta[X] \longleftrightarrow \frac{1}{n}\sum_{i=1}^{n} x_i.$$

This approach is often simple and gives closed-form estimators. However, it has an important limitation: it uses only selected summaries of the data. Two different parameter values may produce similar low-order moments even if they assign very different likelihoods to the observed dataset. A more general idea is therefore:

> Choose the parameter value under which the observed data are most likely.

Suppose

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} p_\theta(x),$$

and we observe the realizations

$$x_1, \ldots, x_n.$$

We need to estimate the parameters θ. The **likelihood function** is the joint probability density of the observed dataset viewed as a function of θ:

$$L(\theta) = p_\theta(x_1, \ldots, x_n).$$

For i.i.d. observations, the joint density can be factorized as

$$\boxed{L(\theta) = \prod_{i=1}^{n} p_\theta(x_i).}$$

The **maximum likelihood estimator** is then

$$\boxed{\widehat{\theta}_{\mathrm{MLE}} = \arg\max_\theta L(\theta).}$$

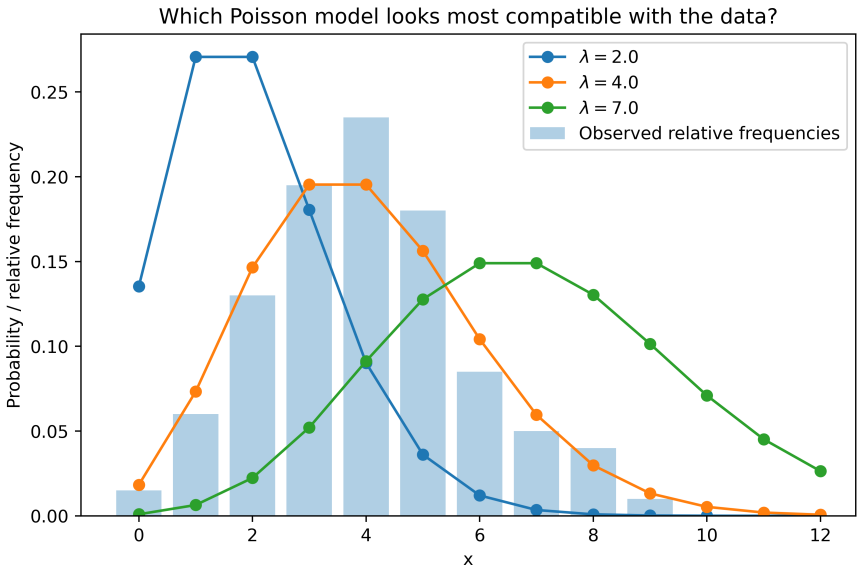


Figure 1: Observed relative frequencies from a finite sample drawn from a Poisson distribution, compared with candidate Poisson models with different values of λ. Maximum likelihood chooses the value of λ that makes the observed sample most likely. NOTE: Because the dataset is finite, the empirical distribution does not exactly match the true population distribution.

Thus, parameter estimation becomes an optimization problem.
In practice, we usually maximize the **log-likelihood**

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log p_\theta(x_i),$$

because the logarithm converts products into sums. Since $\log(\cdot)$ is strictly increasing,

$$\boxed{\arg\max_\theta L(\theta) = \arg\max_\theta \ell(\theta).}$$

Equivalently, one can minimize the negative log-likelihood,

$$\boxed{\widehat{\theta}_{\text{MLE}} = \arg\min_\theta \big[-\ell(\theta) \big],}$$

which is the form commonly used in numerical optimization and machine learning.

2.8 Example: Poisson distribution

Suppose

$$X_i \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda), \qquad p_\lambda(x) = \frac{\lambda^x e^{-\lambda}}{x!}.$$

Given observed values $x_1, \ldots, x_n$, the likelihood is

$$L(\lambda) = \prod_{i=1}^{n} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}.$$

Taking log,

$$\ell(\lambda) = \sum_{i=1}^{n} \left[x_i \log \lambda - \lambda - \log(x_i!) \right].$$

Differentiate with respect to λ:

$$\frac{d\ell}{d\lambda} = \frac{1}{\lambda} \sum_{i=1}^{n} x_i - n.$$

Setting the derivative equal to zero gives

$$\widehat{\lambda}_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}_n.$$

Hence,

$$\boxed{\widehat{\lambda}_{\text{MLE}} = \bar{x}_n.}$$

For the Poisson distribution, the maximum likelihood estimator happens to coincide with the method-of-moments estimator. This agreement does not hold in general.

2.9 Maximum likelihood estimation for a Gaussian distribution

Suppose

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2),$$

where both μ and σ^2 are unknown. The Gaussian density is

$$p(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right].$$

Given observed values $x_1, \ldots, x_n$, the likelihood is

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x_i-\mu)^2}{2\sigma^2}\right].$$

Taking log,

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2.$$

The maximum likelihood estimates are obtained by solving

$$\frac{\partial \ell}{\partial \mu} = 0, \qquad \frac{\partial \ell}{\partial \sigma^2} = 0.$$

For the mean,

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) = 0 \implies \hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

so

$$\boxed{\widehat{\mu}_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}_n.}$$

Similarly, for the variance,

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(x_i-\mu)^2.$$

Substituting $\widehat{\mu}_{\text{MLE}} = \bar{x}_n$ and setting the derivative equal to zero gives

$$\boxed{\widehat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x}_n)^2.}$$

Thus, for Gaussian data,

$$\boxed{\widehat{\mu}_{\text{MLE}} = \bar{x}_n, \qquad \widehat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x}_n)^2.}$$

These are the same estimators obtained earlier using the method of moments. Note that

$$\widehat{\sigma}^2_{\text{MLE}} \neq S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x}_n)^2,$$

so the MLE of Gaussian σ^2 is not the usual unbiased sample variance.